# VECTOR FIELDS ON SMOOTH PROPER CURVES: VANISHING IN GENUS AT LEAST 2, FIXED-POINT COUNTS IN GENUS 0 AND 1

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the derivation identification, the degree computations in all three genera, and the uniformizer-based fixed-point translation are all retained verbatim.

*Editorial note.* The label lemma-smooth-vector-fields names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Derivations as twisted differentials. Same chapter, Lemmas lemma-duality-dim-1 and lemma-uniformizer-works. Identifies Der_k(O_X, O_X) with Hom(omega_X, O_X) via D = theta composed with d, and makes dz a basis of Omega at a closed point for a uniformizer z.
- Riemann-Roch degree formula. Same chapter, Lemma lemma-rr. Gives deg(omega_X) = 2g - 2, the input to the vanishing and vanishing-locus computations.
- Global sections of invertible sheaves on curves. Varieties, Lemma varieties-lemma-check-invertible-sheaf-trivial. Forces theta = 0 when Hom(omega_X, O_X) = H^0(omega_X^tensor -1) has negative degree, and controls the vanishing divisor in the genus 0 and 1 cases.

*Source reference.* Original reference 0BS2: sources/stacks-curves/pinned-original-source.tex, source lines 796–796. Prerequisite pointer invoked at proof line(s) 6897; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0BS6: sources/stacks-curves/pinned-original-source.tex, source lines 1218–1218. Prerequisite pointer invoked at proof line(s) 6899; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0C1E: sources/stacks-curves/pinned-original-source.tex, source lines 2629–2629. Prerequisite pointer invoked at proof line(s) 6911; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0B40: sources/stacks-chow/tags.txt, source lines 13983–13983. Prerequisite pointer invoked at proof line(s) 6902; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $k$ be an algebraically closed field. Let $X$ be a smooth, proper, connected curve over $k$. Let $g$ be the genus of $X$.*

(1) *If $g \geq 2$, then $Der_k(\mathcal{O}_X, \mathcal{O}_X)$ is zero,*
(2) *if $g = 1$ and $D \in Der_k(\mathcal{O}_X, \mathcal{O}_X)$ is nonzero, then $D$ does not fix any closed point of $X$, and*
(3) *if $g = 0$ and $D \in Der_k(\mathcal{O}_X, \mathcal{O}_X)$ is nonzero, then $D$ fixes at most 2 closed points of $X$.*

*Proof.* Recall that we have a universal $k$-derivation $d : \mathcal{O}_X \to \Omega_{X/k}$ and hence $D = \theta \circ d$ for some $\mathcal{O}_X$-linear map $\theta : \Omega_{X/k} \to \mathcal{O}_X$. Recall that $\Omega_{X/k} \cong \omega_X$, see Lemma 0BS2. By Riemann-Roch we have $\deg(\omega_X) = 2g - 2$ (Lemma 0BS6). Thus we see that $\theta$ is forced to be zero if $g > 1$ by Varieties, Lemma 0B40. This proves part (1). If $g = 1$, then a nonzero $\theta$ does not vanish anywhere and if $g = 0$, then a nonzero $\theta$ vanishes in a divisor of degree 2. Thus parts (2) and (3) follow if we show that vanishing of $\theta$ at a closed point $x \in X$ is equivalent to the statement that $D$ fixes $x$ (as defined above). Let $z \in \mathcal{O}_{X,x}$ be a uniformizer. Then $dz$ is a basis element for $\Omega_{X,x}$, see Lemma 0C1E. Since $D(z) = \theta(dz)$ we conclude. $\square$